A High-Reliable and High-Speed All-Digital STT-MRAM-Based Computing-in-Memory for Binary Neural Networks

Yongcheng Wang[1], Tao Li[1,2], and Tetsuo Endoh[1,2], *Fellow, IEEE*

[1]School of Engineering, Tohoku University, Sendai 980-8577, Japan
[2]Center for Innovative Integrated Electronic Systems, Sendai 980-0845, Japan

Computing-in-memory (CiM) has been regarded as a promising AI accelerator architecture owing to its high-energy efficiency and high throughput. This work proposes a novel spin-transfer torque magnetic random access memory (STT-MRAM)-based CiM architecture for binary neural networks (BNNs) to address the limitations in analog STT-MRAM-based CiM designs, which suffer from limited tunneling magnetoresistance (TMR) in magnetic tunnel junctions (MTJs). The key improvements to the proposed architecture are as follows: 1) accelerated binary multiplication via a differential-type architecture; 2) improved tolerance to TMR through a digital multiplication readout scheme; and 3) reduced power consumption owing to the high-speed computation. SPICE simulation shows that the proposed CiM cell can achieve a 0.68 ns low computing delay at a supply voltage of 1 V and 10 fF output parasitic capacitance, and three times increase in output swing at a TMR of 50%.

***Index Terms*— Binary neural networks (BNNs), digital computing-in-memory (DCiM), high-reliable and high-speed, spin-transfer torque magnetic random access memory (STT-MRAM).**

I. Introduction

COMPUTING-IN-MEMORY (CiM) is a hot research topic now for power-efficient AI accelerator design. The realization of efficient CiM systems strongly depends on the underlying memory technology. magnetic random access memory (MRAM) has nonvolatility, long time retention, unlimited write endurance, etc. [1], making it a compelling candidate for next-generation memory and CiM platforms.

Several prior works have explored MRAM-based CiM architectures [2], [3], [4], [5], [6], [7], [13], [15], [16], including spin-orbit torque magnetoresistive random access memory (SOT-MRAM) and spin-transfer torque MRAM (STT-MRAM). SOT-MRAM has better properties but is not mature to fabricate. The more mature STT-MRAM-based CiM is mainly by leveraging the analog behavior of magnetic tunnel junctions (MTJs). In these designs, analog MAC operations are implemented using current-mode sensing or charge accumulation across bitlines or capacitive nodes, thus needs a modified readout sensing circuit design. However, analog CiM has serious drawbacks: 1) low tunneling magnetoresistance (TMR) ratio of MTJs, typically less than 200%, limits the distinguish ability of resistance states during sensing and 2) large energy and area overhead due to the complicated sense amplifier (SA) or analog-to-digital converter (ADC) readout mechanisms [7], [14], [15].

At the algorithm level, binary neural networks (BNNs), in which both weights and input activations are binarized to "+1" and "−1" [8], have emerged as ideal candidates for resource-constrained computing due to their lightweight nature [13]. However, the foundry-compatible 1-transistor-1-MTJ (1T1M) STT-MRAM cell faces limitations in realizing binary multiplication. As a result, new cell structures are designed to utilize the binary dot-product operation.

To overcome these limitations, this work proposes an all-digital CiM architecture based on STT-MRAM. In the proposed architecture, the computation is implemented in a digital computing method, eliminating the need for analog sensing, current summation, or charge-based computation. Bitwise XNOR operations between stored weights and input activations are performed using digital peripheral logic. In brief, the main contributions of this article are summarized as follows.

1. We propose an all-digital STT-MRAM-based CiM cell, supporting high-speed and high-reliable in-memory binary multiplication without relying on any SA or ADC readout circuits.

2. We develop four types of CiM cell structures to realize the digital CiM architecture, comparing the reliability and power performances of them and select the most suitable one for CiM.

3. We present circuit-level design and simulation results, validating the functionality and performance of the proposed architecture under practical device parameters and operational constraints, and compare the reliability of TMR, process corner and parasitic parameters, computing delay and power consumption with the previous work in [7].

This work is an extended version of our previous work [9]. The rest of this article is organized as follows. Section II briefly introduces the architectures of the digital CiM cells. The simulation results and benchmarking are given in Section III. Finally, Section IV concludes this article.

II. Proposed STT-MRAM-Based Digital CiM Architecture

To enable digital-style computing-in-memory (CiM), a common design approach is to integrate logic computation circuits

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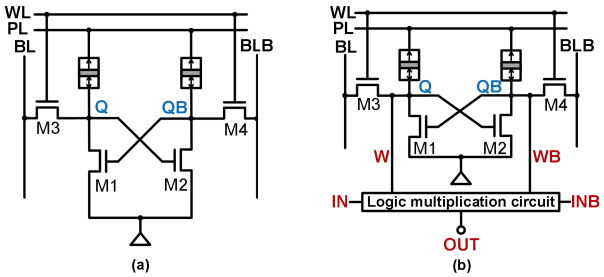


Fig. 1. (a) Schematic of 4T2M differential type STT-MRAM cell [12]. (b) Circuit structure of the digital STT-MRAM-based CiM cell.

with memory cells [10], [11]. However, this approach is difficult to implement on foundry 1T1M STT-MRAM cell structures due to the lack of latch-like configurations as found in static random access memory (SRAM). Instead, such designs rely on SA or ADC for data readout, which leads to degraded computational accuracy and speed.

To address this issue, the proposed design adopts a 4-transistor-2-MTJ (4T2M) differential type STT-MRAM [12]-based digital CiM cell structure, as shown in Fig. 1(a). This structure incorporates a latch-like mechanism similar to that of SRAM, enabling the stored data in MTJs to be rapidly restored to the storage nodes upon activation. As a result, SAs or ADCs are not required for computation, which enhances both speed and accuracy in digital CiM operations.

The proposed CiM architecture, depicted in Fig. 1(b), comprises the memory and the logic multiplication. The memory adopts a 4T2M differential type STT-MRAM configuration. For neural networks processing, weight data are written into the MTJs via BL/BLB lines. During standby periods, the power line (PL) is power gated to minimize leakage. Upon activation, the PL restores stored weight data from the MTJs to the internal nodes Q and QB, which controls transistors of the multiplication circuit. The activation inputs IN and INB subsequently determine the state of the multiplication result OUT, which either remains high or low voltages, representing logical "1" or "0," respectively. The results will be outputted directly to the logic adder trees outside of the CiM cells to perform the addition computation.

Fig. 2 illustrates the circuit implementations of four types of CiM cells based on the proposed digital CiM architecture. The upper part of each CiM cell employs a 4T2M differential type STT-MRAM structure for data storage, while the lower part performs multiplication operations using different circuit configurations: 1) two nMOS transistors; 2) two pMOS transistors; 3) two CMOS transmission gates; and 4) two nMOS transistors along with an external charging branch. In configurations 1)–3), the output nodes remain at a low voltage level when inactive. In contrast, the output of configuration 4) is charged by an external charging circuit during the idle state, maintaining a high voltage level when inactive. For convenience of description, these four cells are hereafter referred to as nMOS-type, pMOS-type, TG-type, and charge-type, respectively.

In BNN, the binary multiplication between weights and activations corresponds to the truth table of the XNOR logic

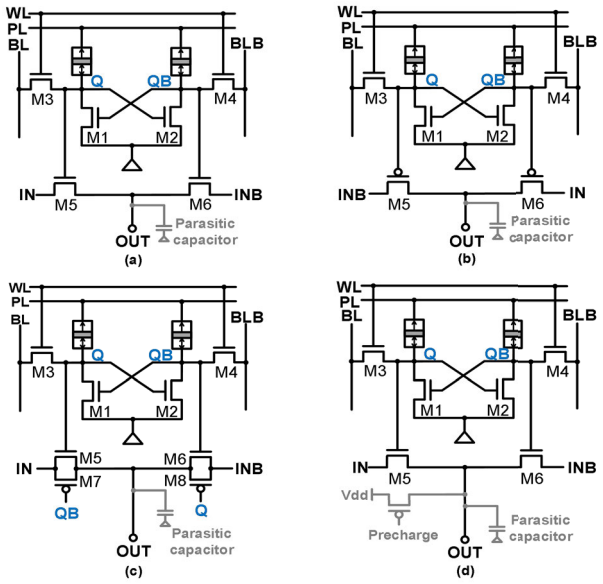


Fig. 2. Schematics of the proposed digital CiM cells: (a) nMOS-type: using 2 nMOS as the logic multiplication and (b) pMOS-type: using 2 pMOS as the logic multiplication. (c) TG-type: using 2 CMOS transmission gates as the logic multiplication. (d) charge-type: using 2 nMOS and an external charge circuit as the logic multiplication.

TABLE I
BINARY MULTIPLICATION AND XNOR OPERATION IN CIM CELLS

Weight	Input activation	Result
Q/QB	IN/INB	OUT
0/1	0/1	1
0/1	1/0	0
1/0	0/1	0
1/0	1/0	1

operation, as shown in Table I, where in circuit-level implementation "−1" can be encoded as logic "0." Therefore, performing an XNOR operation in hardware is functionally equivalent to executing a binary multiplication between the weight and activation values. In the four CiM cells presented in Fig. 2, the lower computation part serves as an XNOR logic unit, which performs the binary multiplication between the stored node value (Q) and the external input (IN).

III. EVALUATION AND BENCHMARKING

A. Simulations of the Proposed Four Types of CiM Cells

All four proposed digital CiM cells can perform XNOR logic operations, which correspond to binary multiplication between neural network weights and input activations. However, due to differences in circuit configurations, their computational performance may vary. To validate the functionality of these CiM cells and compare their performances to identify the most suitable candidate for further research, circuit-level simulations are conducted using the SPICE simulator.

The simulation is implemented using a 55 nm process technology. An MTJ model, characterized by a nominal TMR of 122% and an ambient temperature of 25 °C, is used for simulation. All the width of nMOS transistors of CiM cells

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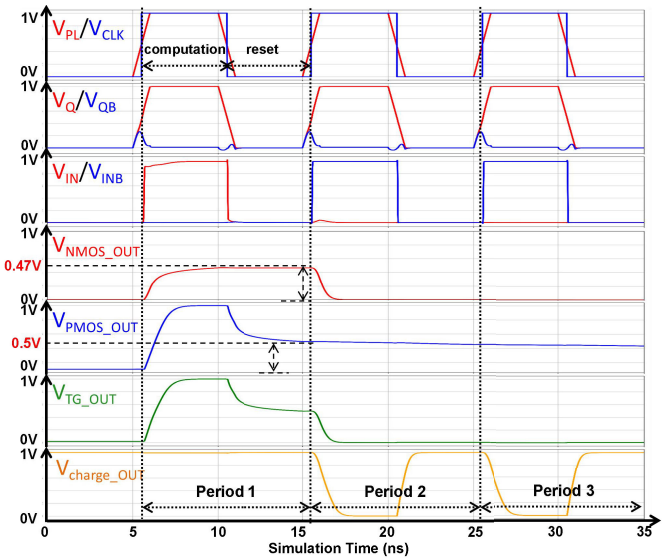


Fig. 3. Simulation results of four digital CiM cells when input pattern is 1-0-0.

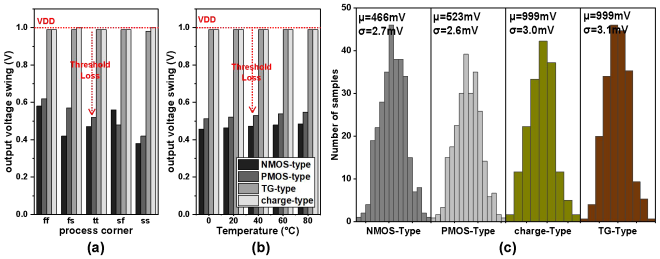


Fig. 4. Simulation results of four CiM cells' output voltage swing under different (a) process corner, (b) temperature, and (c) 3σ variation of supply voltage.

are set to 0.3 μm and pMOS to 0.6 μm, a capacitor at the OUT node is added to simulate the parasitic capacitance of output line in the real array, and in this simulation to highlight the results more clearly, the capacitor is set to 100 fF.

Fig. 3 illustrates the waveforms of various signals during the computation cycles under a supply voltage of 1 V. Each operation cycle spans 10 ns, comprising 5 ns for computation and 5 ns for reset. During the reset period, all signal activities are halted. In all simulated CiM cells, the two MTJs are configured such that the left MTJ is in the low-resistance state and the right MTJ is in the high-resistance state, effectively encoding a stored weight value of "1." To verify the computation, the input signals IN and INB are driven with a serial 3-bit sequence, which needs three operation cycles. In Fig. 3, the input pattern is "1-0-0" in a logic representation.

When the input pattern is "1-0-0" and the restored voltage states of Q/QB are "1/0," the expected computation results are "1-0-0," respectively. As shown in Fig. 3, under a 5 ns computation window, all four CiM cells successfully generate the corresponded output values within the computation period, with delays even shorter than 5 ns. However, for nMOS-type and pMOS-type CiM cells, the output voltage suffers from significant threshold voltage degradation.

According to the digital computing-in-memory (DCiM) architecture, the computation results of the CiM cells are directly delivered to the adder tree for accumulation. Therefore, the output voltage level has a significant impact on the correctness of the adder. To show the reliability of computation, SPICE simulations of the four CiM cells' output voltage swing are carried out under different process corners and temperatures ranging from 0 °C to 80 °C, as well as a Monte Carlo simulation with voltage variations of 3σ, as shown in Fig. 4(a)–(c), respectively. Fig. 4(a) shows that when TG and charge-type CiM cells have a better reliability to process corner variation, whereas nMOS and pMOS-type CiM cells are sensitive to it. Fig. 4(b) and (c) shows all CiM cells have a good reliability to temperature and voltage variations. However, the swing of the nMOS and pMOS-type CiM cells cannot reach the full range due to the threshold voltage drop. This limitation may lead to logic errors and propagation failures in subsequent computations, thereby posing a significant threat to computational accuracy. On the other hand, the TG and charge-type CiM cells effectively overcome the threshold voltage degradation issue by introducing additional circuit elements. These enhancements enable full-swing output behavior between VDD and GND under different variations, which is essential for robust digital operation. However, the inclusion of additional devices leads to increases in cell area and power consumption. Therefore, to evaluate the trade-offs between these two improved CiM cell designs, a more detailed performance comparison is required.

B. Power Comparison of TG and Charge-Type CiM Cells

To investigate the power consumption characteristics of the TG-type and charge-type CiM cells, it is necessary to analyze the sources of power consumption. During operation, the total power consumption primarily originates from two sources: 1) the current flowing from the PL to ground through the upper latch structure when PL is active and 2) the dynamic charging and discharging of the parasitic capacitance at the output node (OUT) in the lower circuit during binary multiplication. Since the CiM cell does not consume static power in the idle state, and the static power during the computation period is negligible compared to dynamic power, static power consumption is omitted from the analysis. Due to the structural separation of the storage and computation logic in the proposed digital CiM cell, the current paths of the memory and multiplication are independent, and thus their power consumption can be analyzed separately.

For different input patterns, the power consumed due to PL activation remains constant, as it only depends on the supply voltage and the duration of PL being enabled during a computation cycle. In contrast, the dynamic power resulting from the charging and discharging of the OUT node's parasitic capacitance varies with different computation results. Fig. 5(a) presents the output current waveforms of the two CiM cells under the input patterns of "1-0-0" and "1-0-1" as examples of the eight types of input patterns. In these waveforms, positive current indicates charging of the output parasitic capacitance, while negative current corresponds to discharging behavior.

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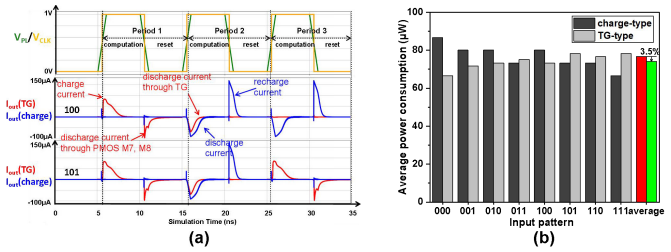


Fig. 5. (a) OUT current waveforms of TG type and charge type CiM cells under the eight types of input patterns. (b) Average power of three computations of the two types of CiM cells under the eight types of input pattern.

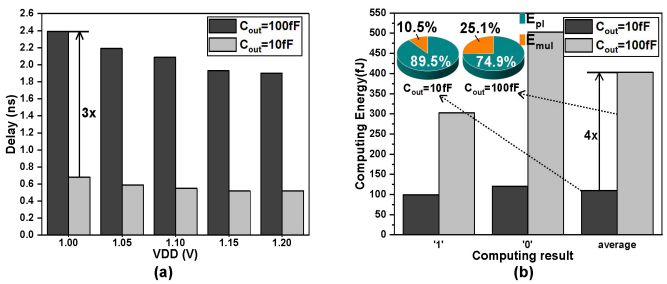


Fig. 6. (a) Computation delay under varying supply voltages when output parasitic capacitance is 10 and 100 fF, respectively. (b) Computation power of the proposed CiM cell at a supply voltage of 1 V with output parasitic capacitances of 10 and 100 fF, respectively.

As observed from the simulation results, for the charge-type CiM cell: when the computation result is "1," no charging or discharging occurs at the output node, since the external power supply continuously charges the parasitic capacitance during the idle state. Consequently, no output current is generated. When the computation result is "0," the parasitic capacitance at the output node discharges through the IN/INB port, producing a discharge current. This is followed by the reset phase, during which the capacitance is recharged by the external supply. For the TG-type CiM cell: when the computation result is "1," the IN/INB port charges the parasitic capacitance, resulting in a charging current. During the subsequent reset phase, as the CiM cell is unpowered and the Q/QB node voltages drop to 0 V, the two pMOS transistors (M7 and M8) within the transmission gate become active. The charge stored in the parasitic capacitor during computation then leaks through the pMOS transistors to the IN/INB ports, generating a discharge current. However, since TGs are not fully turned on under these conditions, the discharge occurs slowly, and the parasitic capacitance is not completely discharged. If the next computation result is "0," the TG is fully turned on, allowing the remaining charge to be released, which produces another discharge current. If the parasitic capacitance holds no charge at the beginning of a computation cycle, no charging or discharging current will occur.

Fig. 5(b) compares the average power consumption of the two CiM cells over three computation cycles under eight different input patterns. In terms of total average power across all eight input patterns, the TG-type CiM cell exhibits slightly lower power consumption, approximately 3.5% less than that of the charge-type CiM cell. However, this reduction comes at the cost of increased cell area and slower charge/discharge behavior. Therefore, from an overall performance perspective, the charge-type CiM cell is considered more suitable for use as a digital CiM cell.

C. Benchmarking

This section gives a more detailed analysis of the proposed charge-type CiM cell including computing delay, energy, and reliability. At the same time, benchmarking with some other STT-MRAM-based CiM are performed. The results are based on the cell-level SPICE simulation; however, to make it realistic, different parasitic capacitance values of the output are used.

Fig. 6(a) shows the XNOR computation delay of the proposed CiM cell under varying supply voltages when output parasitic capacitance is 10 and 100 fF, respectively. The delay is defined as the time required for the output voltage to reach a steady value. As shown in Fig. 6(a), at a 1 V supply voltage and 10 fF parasitic capacitance, the delay is only 0.68 ns. When the capacitance increases from 10 to 100 fF, the delay becomes three times larger. The parasitic capacitance is primarily determined by the number of cells connected to a single output line. In the practical circuit implementation, since the output line is not required to connect multiple cells as in the case of the WL or BL, its parasitic capacitance can be effectively controlled to achieve high-speed computation.

Fig. 6(b) illustrates the computation power of the proposed CiM cell at a supply voltage of 1 V with output parasitic capacitances of 10 and 100 fF, respectively. Unlike the previous power simulations, in this case the PL is immediately turned off once the computation result of the CiM cell is sampled, in order to emulate realistic operation and reduce power consumption. As observed, when the output parasitic capacitance increases from 10 to 100 fF, the average computation power for both logic "0" and "1" results increase by approximately four times. This power increase originates significantly not only from the higher charging and discharging energy of the computation units but also from the extended power-on duration of the memory section. From the energy breakdown in Fig. 6(b), it can be observed that during computation, the power consumption of the memory PL accounts for 89.5% and 74.9% of the total power at parasitic capacitances of 10 and 100 fF, respectively, making it the dominant source of power consumption. Therefore, increasing the computation speed to allow earlier shutdown of the PL is an effective approach to reducing the computation power.

To highlight the high reliability of the proposed CiM cell structure, a conventional differential STT-MRAM-based analog CiM cell structure with a latch SA circuit as shown in [7] is replicated for comparison. Simulations are performed at a 1 V supply voltage and 25 °C. Fig. 7 shows the results of the output voltage swing of the two CiM cell structures when the computation result is either "0" or "1," under three different variations: process corner variation, output parasitic capacitance, and TMR fluctuation. The shaded rectangles indicate the voltage swing range under each condition. As shown

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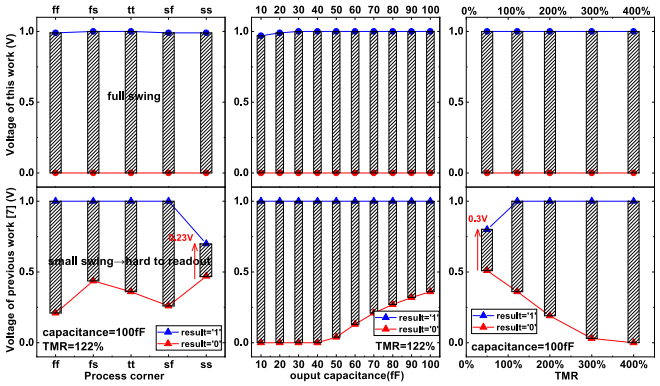


Fig. 7. Output voltages and swing of the previous work and this work when the computation result is "1" or "0" under different process corner, output capacitance, and TMR.

TABLE II
COMPARISON WITH STATE-OF-THE-ART STT-MRAM-BASED CIM

	This work	TCAS-II'23 [2]	TMAG '23 [3]	JETCAS '22 [4]	ISSCC' 23 [7]
Process	55nm	28nm	N/A	65nm	28nm
Memory type	STT-MRAM	STT-MRAM	STT-MRAM	STT-MRAM	STT-MRAM
Input bit	1b	1b	1b	1b	9b
Weight bit	1b	1b	2b	1b	9b
Delay for readout (ns)	0.68[1]	10	6.6	10	12
Energy for per bit computation (fJ/b)	109.9[1,3]	163.7[3]	40.5[3]	6.4[2,4]	57[4]
DCiM or ACiM	DCiM	Mixed CiM	ACiM	ACiM	ACiM

[1]simulation without a peripheral circuit. [2]estimated results according to data given in the paper. [3]Energy for per bit multiplication. [4]Energy for per bit MAC.

in the figure, the proposed CiM cell consistently achieves full-swing output from VDD to GND across all tested conditions. In contrast, the output swing of the benchmark fluctuates significantly with these variations and, in some cases, falls short of even half the full swing. For example, the output swing of the proposed CiM cell is three times larger than the benchmark when TMR is 50%. This demonstrates the good robustness of the proposed design, which benefits from its digital structure and inherent resilience to nonlinearities introduced by device and circuit-level variations.

Table II shows a comparison of this work and some state-of-the-art STT-MRAM-based CiM. The results of this work are simulated at a 10 fF output parasitic capacitance. From this table, it can be observed that although the proposed structure does not exhibit particularly low power consumption, it achieves extremely low computation latency because no additional readout circuitry is required during the result extraction process. This characteristic highlights the advantage of the proposed design in application scenarios that demand ultrahigh-speed computation.

IV. CONCLUSION

A novel all-digital STT-MRAM based CiM architecture has been presented in this work. Through a state-of-the-art comparison, the proposed CiM cell demonstrates a 0.68 ns readout delay, three times larger output swing at a 50% TMR, and full swing under different variations. These advantages underscore its suitability for high-speed and high-reliability for AI accelerator applications.

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